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State	Finished
Completed on	Monday, 12 October 2020, 3:23 PM
Time taken	21 mins 46 secs
Marks	5.00/15.00
Grade	**33.33** out of 100.00

Question **1**

Complete

Mark 0.00 out of 1.00

When do we use a HAVING clause?

Select one:

- **a. To limit the output of a query**
- **b. To limit the output of a query using an aggregate function only**
- **c. When GROUP by is used**
- **d. both To limit the output of a query using an aggregate function and When GROUP by is used**

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following standard algorithms is not a Greedy algorithm?

Select one:

- **a. Prim's algorithm**
- **b. Kruskal algorithm**
- **c. Dijkstra's shortest path algorithm**
- **d. Huffman Coding**
- **e. Bellmen Ford Shortest path algorithm**

Question 3

Complete

Mark 1.00 out of 1.00

Given the following Managers Table

Managers –

Id	Name	Salary
1	Harpreet	20000
2	Ravi	30000
3	Vinay	10000
4	Ravi	30000
5	Harpreet	20000
6	Vinay	10000
7	Rajeev	40000
8	Vinay	10000
9	Ravi	30000
10	Sanjay	50000

what will be the rows of the table after the following query is executed

```
DELETE M1
From managers M1, managers M2
Where M2.Name = M1.Name AND M1.Id>M2.Id;
```

Select one:

a.

Id	Name	Salary
1	Harpreet	20000
2	Ravi	30000
3	Vinay	10000
7	Rajeev	40000
10	Sanjay	50000

b. Managers –

Id	Name	Salary
1	Harpreet	20000
2	Ravi	30000
4	Ravi	30000
5	Harpreet	20000
7	Rajeev	40000
9	Ravi	30000
10	Sanjay	50000

c. Managers –

Id	Name	Salary
1	Harpreet	20000
3	Vinay	10000
5	Harpreet	20000
6	Vinay	10000

/

7	**Rajeev**	**40000**
8	**Vinay**	**10000**
10	**Sanjay**	**50000**

d. Managers –

Id	Name	Salary
1	**Harpreet**	**20000**
2	**Ravi**	**30000**
3	**Vinay**	**10000**
4	**Ravi**	**30000**
5	**Harpreet**	**20000**
6	**Vinay**	**10000**
7	**Rajeev**	**40000**
8	**Vinay**	**10000**
9	**Ravi**	**30000**
10	**Sanjay**	**50000**

Question 4

Complete

Mark 1.00 out of 1.00

Given relations R(w,x) and S(y,z), the result of

SELECT DISTINCT w, x

FROM R, S

Is guaranteed to be same as R, if

Select one:

- **a. R and S have no duplicates**
- **b. R and S have the same number of tuples**
- **c. R has no duplicates and S is non-empty**
- **d. S has no duplicates and R is non-empty**

/

Question 5

Complete

Mark 0.00 out of 1.00

Consider the following C program that attempts to locate an element x in an array Y[] using binary search.

The program is erroneous.

```
1.  f(int Y[10], int x) {
2.    int i, j, k;
3.    i = 0; j = 9;
4.    do {
5.          k =  (i + j) /2;
6.          if( Y[k] < x)  i = k; else j = k;
7.       } while(Y[k] != x && i < j);
8.    if(Y[k] == x) printf ("x is in the array ") ;
9.    else printf (" x is not in the array ") ;
10. }
```

On which of the following contents of Y and x does the program fail?

Select one:

- **a. Y is [2 2 2 2 2 2 2 2 2 2] and x > 2**
- **b. Y is [2 4 6 8 10 12 14 16 18 20] and 2 < x < 20 and x is even**
- **c. Y is [1 3 5 7 9 11 13 15 17 19] and x < 1**
- **d. Y is [1 2 3 4 5 6 7 8 9 10] and x < 10**

Question 6

Complete

Mark 0.00 out of 1.00

Given:

```
3. class Alpha {
4. static String s = " ";
5. protected Alpha() { s += "alpha "; }
6. }
7. class SubAlpha extends Alpha {
8. private SubAlpha() { s += "sub "; }
9. }
10. public class SubSubAlpha extends Alpha {
11. private SubSubAlpha() { s += "subsub "; }
12. public static void main(String[] args) {
13. new SubSubAlpha();
14. System.out.println(s);
15. }
16. }
```

What is the result?

Select one:

- **a. sub subsub**
- **b. alpha sub subsub**
- **c. An exception is thrown at runtime**
- **d. Compilation fails**
- **e. subsub**
- **f. alpha subsub**

Question 7

Complete

Mark 0.00 out of 1.00

```
void f(char *s)
{
char *s1;
char c;
s1=s+(strlen(s)-1);
while(s<s1)
      {
       c=*s;
       *s=*s1;
       *s1=c;
       s++;
       s1--;
      }
printf("%s %s",s,s1);
}

 main()
{
f("hello");
}
```

Select one:

- **a. leh leh**
- **b. None of these**
- **c. hell hell**
- **d. hello hello**
- **e. olleh olleh**

Question 8

Complete

Mark 0.00 out of 1.00

```
3. public class Redwood extends Tree {
4. public static void main(String[] args) {
5. new Redwood().go();
6. }
7. void go() {
8. go2(new Tree(), new Redwood());
9. go2((Redwood) new Tree(), new Redwood());
10. }
11. void go2(Tree t1, Redwood r1) {
12. Redwood r2 = (Redwood)t1;
13. Tree t2 = (Tree)r1;
14. }
15. }
16. class Tree { }
```

What is the result? (Choose all that apply.)

Select one:

- **a. Compilation fails with an error at line 9**
- **b. An exception is thrown at runtime**
- **c. Compilation fails with an error at line 12**
- **d. The code compiles and runs with no output**
- **e. Compilation fails with an error at line 8**
- **f. Compilation fails with an error at line 13**

Question 9

Complete

Mark 0.00 out of 1.00

Which of the following sorting algorithms in its typical implementation gives best performance when applied on an array which is sorted or almost sorted

(maximum 1 or two elements are misplaced).

Select one:

- **a. Heap Sort**
- **b. Merge Sort**
- **c. Insertion Sort**
- **d. Quick Sort**

Question 10

Complete

Mark 0.00 out of 1.00

```
void main()
{
 int a=0xff;
 if(a<<4>>12)
        printf("zemoso");
 else
        printf("labs");
}
```

Select one:

- **a. zemoso**
- **b. None of these**
- **c. Compilation error**
- **d. labs**

Question 11

Complete

Mark 1.00 out of 1.00

```
4. public class Frodo extends Hobbit {
5. public static void main(String[] args) {
6. Short myGold = 7;
7. System.out.println(countGold(myGold, 6));
8. }
9. }
10. class Hobbit {
11. static int countGold(int x, int y) { return x + y; }
12. }
```

What is the result?

Select one:

- **a. Compilation fails due to an error on line 6**
- **b. Compilation fails due to multiple errors**
- **c. Compilation fails due to an error on line 7**
- **d. Compilation fails due to an error on line 11**
- **e. 13**

Question 12

Complete

Mark 1.00 out of 1.00

Statment 1 :- (INNER) JOIN: Returns records that have matching values in both tables

Statment 2 :- LEFT (OUTER) JOIN: Return all records from the left table, and the matched records from the right table

Statment 3 :- RIGHT (OUTER) JOIN: Return all records from the right table, and the matched records from the left table

Statment 4 :- FULL (OUTER) JOIN: Return all records when there is a match in either left or right table

Select one:

- **a. Statement 1,2 is true, Statement 3,4 are false**
- **b. Statement 1,2,3,4 are all true**
- **c. Statement-1 is true, Statement 2,3,4 are false**
- **d. Statement 1,2,3 is true , Statement 4 is false**

Question 13

Complete

Mark 0.00 out of 1.00

```
import java.util.*;
class Test {
public static void main(String[] args) {
// insert code here
x.add("one");
x.add("two");
x.add("TWO");
System.out.println(x.poll());
}
}
```

Which, inserted at // insert code here, will compile? (Choose all that apply.)

Select one or more:

- **a. TreeSet<String> x = new TreeSet<String>();**
- **b. List<String> x = new LinkedList<String>();**
- **c. LinkedList<String> x = new LinkedList<String>();**
- **d. HashSet<String> x = new HashSet<String>();**
- **e. Queue<String> x = new PriorityQueue<String>();**
- **f. ArrayList<String> x = new ArrayList<String>();**

Question **14**

Complete

Mark 0.00 out of 1.00

What is the value printed by the following C program?

```
#include<stdio.h>
int f(int * a, int n)
{
if (n<=0)return 0;
else if(*a%2==0)
return *a+f(a+1,n-1);
else return * a-f(a+1, n-1);
}
int main ()
{
int a[]={12, 7, 13, 4, 11, 6};
printf("%d",f(a,6));
return 0;
}
```

Select one:

- **a. 5**
- **b. 15**
- **c. 19**
- **d. -9**

Question **15**

Complete

Mark 0.00 out of 1.00

```
class Top
{
public Top(String s)
{
System.out.print("B");
}
}

public class Bottom2 extends Top
{
public Bottom2(String s) { System.out.print("D"); }
public static void main(String [] args) {
new Bottom2("C");
System.out.println(" ");
} }
```

What is the result?

Select one:

- **a. BD**
- **b. BDC**
- **c. DB**
- **d. DBC**
- **e. Compilation fails**

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